What Makes a Memory Intrusive?

Large Language Model Analysis of Semantic Features in the Trauma Film Paradigm

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Abstract

The Trauma Film Paradigm (TFP) is widely used to study intrusive memories, but what makes an experience elicit intrusive memories remains an open question. Using LLM-based semantic analysis, we examined how features of film segments shape voluntary versus involuntary recall. Crucially, the set-up of the test was matched: in both conditions, participants performed a vigilance task which included presentation of blurred scenes from previously seen negative or neutral clips. In a between-subject design, participants either voluntarily recalled film details or reported when they had an involuntary memory of the films. The semantic analysis code was successful in identifying and assigning the memories to their respective clips and then computing the number of times a reported memory matched the cue. Although there was no difference between the number of memories reported, matches were more frequent in the negative and involuntary conditions, suggesting that they arise more frequently in response to sensory cues, in support of the Dual Representation Account. When considering the quality of memories recalled voluntarily or involuntarily, the level of arousal extracted from LLM models while watching the film segment that preceded the hotspot, the most arousing film segment, was the strongest predictor of negative involuntary memories. Contrastingly, more agentic film segments had a higher likelihood of being recalled voluntarily. These findings highlight the existence of a cognitive anticipatory buildup mechanism that drives intrusive encoding, and they demonstrate the promise of computational methods for advancing models of trauma memory.

1. Introduction

1.1 Background on Trauma Memory

The Dual Representation Account (Brewin et al., 2010; DRA) proposes two routes for encoding two different components of memory. The affective and sensory components, Sensory Representations (S-Reps), are encoded through a network comprising the amygdala, insula and early sensory cortical and subcortical areas. Personal and spatial contexts and structural descriptions, Contextual Representations (C-Reps), are encoded through a network comprising hippocampal, parahippocampal and other structures in the medial temporal lobe, which implement associative binding. Upon intense emotional arousal during a traumatic event, the network encoding S-Reps is up-modulated and the network encoding C-Reps is down-regulated. This results in ineffective integration of the affective and sensory elements with the spatiotemporal context, resulting in a fragmented episodic memory (Brewin et al., 2010). This provides a cognitive basis for the triggering of unwanted involuntary memories when faced with sensory cues similar to aspects of a traumatic event, even in spatiotemporally dissimilar contexts. In sum, DRA asserts that there are two memory traces, S-Reps and C-Reps, and the encoding of trauma strengthens the former and weakens the latter, leading to dissociations between involuntary and voluntary memory.

The DRA finds grounding in clinical evidence indicating that intrusive memories in Post-Traumatic Stress Disorder (PTSD, American Psychiatric Association, 2013) are often fragmented (Brewin et al., 1996; McNally et al., 2022). In this context, "fragmented" refers to traumatic memories being recalled as disjointed, sensory-heavy fragments lacking coherent narrative structure or integration into autobiographical memory. One method to study the encoding of intrusive memories in a controlled environment is to use emotionally distressing films. In this "trauma film" paradigm, participants encode film clips that depict scenes with traumatic content, such as "actual or threatened death, serious injury, or sexual violence", following from the DSM-5 definition of trauma (American Psychiatric Association, 2013), under conditions that encourage maximal immersion. A substantial body of empirical research demonstrates that healthy participants frequently experience involuntary recollections of film clip segments in the days following viewing, as documented through daily diary methods (James et al., 2015, 2016; Kobelt et al., 2024). Similar patterns are also observed in Vigilance Retrieval Tasks (VRTs), a method of obtaining involuntary memories in the laboratory (James et al., 2015, 2016; Lau-Zhu et al., 2019, 2021). The presence of involuntary memories suggests that a similar mechanism of encoding takes place after watching the films as would have occurred post-experiencing an actual

traumatic event. This offers promise that the trauma film paradigm can act as a laboratory model of exposure to trauma and subsequent intrusive memories of the experimental trauma, e.g. in PTSD (James et al., 2016).

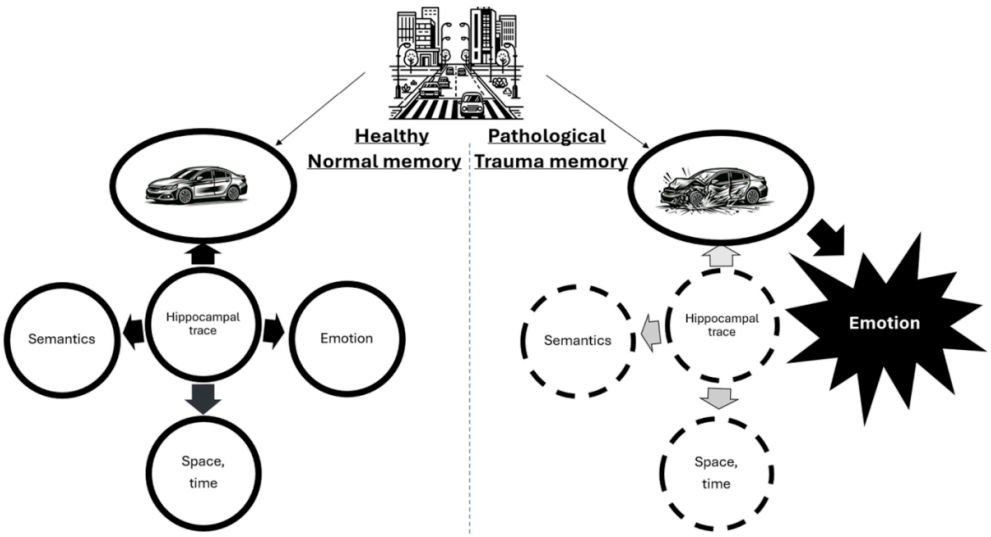


Figure 1: Schematic Representation of DRA of Intrusive Memory (Bernhard et al., 2025)

1.2. Interventions for Intrusions

The Dual Representation Account's core idea, positing two independent memory traces for encoding, contrasts with the unitary account of memory that proposes that a single hippocampal index binds together all details of a memory, both spatio-temporal and multi-modal sensory (Frankland & Bontempi, 2005). Visuo-spatial interventions have come to be investigated as strong evidence for the DRA. One of the most extensively studied approaches involves taxing the visuospatial sketchpad shortly after trauma exposure. Holmes et al. (2009) demonstrated that playing the visuospatial computer game Tetris within the consolidation window significantly reduced the number of intrusive images reported in the days following a trauma film. Crucially, these effects were contingent on the use of reminder cues, such as a still image or a short clip from the film, to bring the memory back into a labile state and disrupt its consolidation or restorage. The proposed mechanism is that competing visuospatial demands disrupt the stabilisation of S-reps, thereby limiting later involuntary access. Follow-up studies have replicated these findings and extended them to clinical analogues, showing that Tetris administered after real-life traumatic events (e.g., emergency department admissions) was associated with fewer subsequent intrusions (Iyadurai et al., 2018). These interventions do not appear to impair deliberate recall of the traumatic material (Lau-Zhu et al., 2019, 2021), consistent with the DRA's distinction between sensory and

contextual representations.

Alternative interventions have targeted verbal processing. Tasks such as word games or podcasts primarily load the phonological loop, but findings suggest they are less effective at reducing intrusions compared to visuospatial tasks (James et al., 2015; Lau-Zhu et al., 2019). This dissociation underscores the modality-specific nature of sensory memory consolidation and suggests that disrupting verbal resources alone may not sufficiently interfere with imagery-based S-reps. Recent findings also raise important caveats. For example, in some versions of the trauma film paradigm, visuospatial tasks like Tetris did not change whether memories came to mind voluntarily or involuntarily, although they did appear to lower the overall number of recollections (Holmes et al., 2009; Iyadurai et al., 2018). This nuance suggests that such tasks influence the encoding of intrusive memories themselves, rather than the manner in which they are later recalled. The finding that visuospatial tasks reduce intrusive memories while leaving voluntary recall intact does not imply a shift in recall modality. Rather, it suggests that such tasks disrupt the formation of intrusive sensory fragments during consolidation, thereby lowering the likelihood of later involuntary intrusions, while leaving narrative, voluntarily accessible memory representations unaffected. Collectively, the literature supports the potential of targeted cognitive tasks as "cognitive vaccines" against intrusive memories, while also emphasising the need for further refinement and translation into real-world contexts.

1.3. NLP in Emotion & Memory Research

The past decade has seen an increasing integration of natural language processing (NLP) methods into cognitive and affective science, particularly for studying memory and emotion. Language models offer a scalable way to quantify semantic and emotional properties of memory descriptions, enabling researchers to test theoretical predictions about what makes certain experiences more likely to intrude. Traditional affective science has long relied on human-rated norms, such as the *Affective Norms for English Words* (ANEW; Bradley & Lang, 1999), which provide standardised ratings of valence (pleasantness) and arousal (physiological activation) for large word sets. While such resources remain foundational, they are limited in scope and cannot fully capture the complexity of naturalistic memory narratives.

Recent advances in distributed representations have provided tools for richer semantic and affective modelling. Word embeddings (e.g., Word2Vec, GloVe) and sentence embeddings (e.g., Sentence-BERT; Reimers & Gurevych, 2019) encode linguistic items into high-dimensional vector spaces where semantic similarity can be quantified via measures such as cosine similarity. These methods allow researchers to automatically map

memory descriptions to their closest originating film segments, offering objective indices of cue-memory overlap (Kobelt et al., 2024). Transformer-based large language models (LLMs) further extend this by extracting nuanced features of narratives, including emotional tone, agency, and temporal focus, with high reliability.

Emotion classification using LLMs has shown promise in capturing dimensions central to trauma memory research. Models such as DistilRoBERTa have been adapted for emotion recognition (Hartmann, 2022), while other classifiers can assess valence or perceived agency in narratives (Wan et al., 2023). These computational measures often correlate with self-reports, though discrepancies highlight the need for careful validation (Barbieri et al., 2020). This offers a powerful alternative for the extraction of semantic features from textual descriptions like agency, valence, and arousal, through more efficient LLM-based pipelines. In the trauma film paradigm, such tools could help in identifying "hotspot" segments, moments of heightened arousal or diminished agency, that disproportionately intrude involuntarily (James et al., 2016). While limitations remain, such as biases in training corpora and reduced sensitivity to non-verbal imagery (Wegge & Klinger, 2024; Xu et al., 2025), NLP methods represent a powerful emerging toolkit for advancing the study of trauma, emotion, and memory.

1.4. Gap & Rationale

While the trauma film paradigm has been widely used to study intrusive memories, most prior studies have relied on manual coding, self-report diaries, or experimental manipulations to understand why certain moments intrude disproportionately (James et al., 2016; Kobelt et al., 2024). Although such approaches provide valuable insights, they are resource-intensive and often limited in scalability. Moreover, the precise characteristics that make a given moment from a traumatic sequence more likely to intrude remain poorly understood.

In particular, few studies have systematically applied computational methods from NLP to model memory intrusions. The use of semantic embeddings and LLMs enables automated extraction of affective dimensions such as valence, arousal, and agency from narrative descriptions, providing a scalable complement to traditional subjective ratings (Reimers & Gurevych, 2019; Hartmann, 2022; Wan et al., 2023). Yet, this computational toolkit has rarely been leveraged within the trauma film paradigm to test predictions about intrusive memories. Additionally, the concept of "hotspots", emotionally intense or perceptually salient segments that disproportionately intrude, was first proposed in clinical models (Ehlers & Clark, 2000) and has since been examined empirically through qualitative analyses highlighting their sensory, spatial, and temporal features (Singh et al., 2022; Hoppe et al., 2022). Early machine learning approaches have also attempted to predict

intrusive memories from neuroimaging data (Clark et al., 2014). However, hotspots have yet to be systematically modelled using fine-grained statistical or computational methods applied directly to narrative or film-based data. Prior work has typically assumed that heightened arousal is the primary driver of intrusive memories (Brewin, 2010; Ehlers et al., 2002), but few studies have tested whether semantic properties (e.g., agency or valence) contribute independently or interactively.

The present study addresses these gaps by combining the trauma film paradigm with NLP-based feature extraction and statistical modelling. By embedding both film segment descriptions and participant memory reports into a shared semantic space, we can systematically assess which features predict voluntary versus involuntary recall. This approach offers a novel way to bridge cognitive theory, affective science, and computational modelling, with implications for both basic memory research and clinical interventions.

1.5. Hypotheses

Based on the DRA (Brewin et al., 2010) and prior empirical findings (Holmes et al., 2009; Iyadurai et al., 2018; James et al., 2016), the following hypotheses are proposed:

1. **Emotional valence and arousal:** Negative, high-arousal clips will elicit more involuntary memories than neutral clips.
2. **Hotspot effects:** Pre-hotspot and arousal-intense segments will be disproportionately recalled involuntarily compared to other segments.

2. Methods

2.1. Participants

Participants were healthy adult volunteers recruited through university mailing lists and online advertisements. Inclusion criteria required participants to be fluent in English and have normal or corrected-to-normal vision. Individuals with a self-reported history of psychiatric disorders, neurological conditions, or current use of psychotropic medication were excluded. Participants were randomly assigned to one of eight experimental conditions in a 2 (Film: Negative vs. Neutral) × 2 (Intervention: Tetris vs. Podcast) × 2 (Intentionality: Voluntary vs. Involuntary) between-subjects design. The final sample size after exclusions was N = 240, balanced across groups.

2.2. Materials

Film stimuli: The experimental trauma induction followed the trauma film paradigm (James et al., 2016), using a 12-minute compilation of 11 clips depicting either traumatic (negative) or neutral content. Clips in the negative condition depicted events meeting the DSM-5 definition of trauma, such as "actual or threatened death, serious injury, or sexual violence" (American Psychiatric Association, 2013). Neutral clips depicted emotionally bland everyday events of comparable length. For example, the neutral films had clips of driving and cooking lessons, whereas the negative conditions had clips of car accidents, a documentary on genocide, and surgeries.

Interventions: Following film viewing, participants completed a reminder cue procedure and were then randomly assigned to either a visuospatial or verbal interference task. In the visuospatial condition, participants played *Tetris*, a computer game shown to disrupt sensory-perceptual memory consolidation (Holmes et al., 2009). In the verbal control condition, participants listened to a podcast episode, intended to occupy verbal working memory resources.

Vigilance task: To probe involuntary memory retrieval, participants later completed a go/no-go number detection task while blurred background images were intermittently presented (Lau-Zhu et al., 2019) as illustrated in Figure 2. Background cues included both stills taken from the film participants had viewed and foil images from unrelated clips.

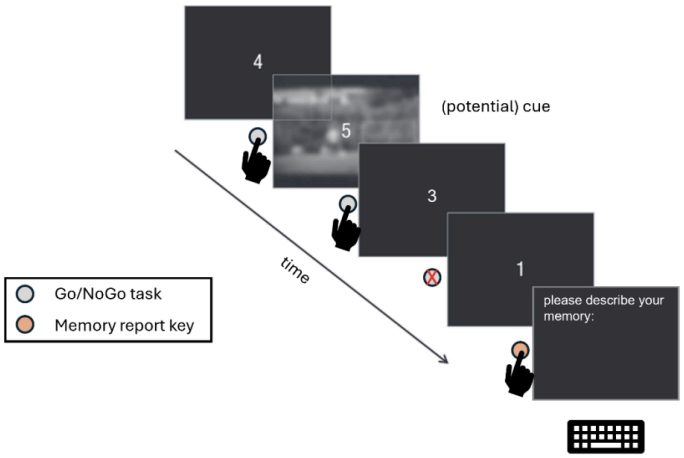


Figure 2: Setup of the Vigilance Retrieval Task (VRT) (Bernhard et al., 2025)

Self-report measures. After the vigilance task, participants rated each film clip on valence (pleasant–unpleasant) and arousal (low–high activation) using standardised Likert scales (Bradley & Lang, 1999).

2.3. Procedure

After providing informed consent, participants filled in a mood rating form and were then randomly assigned to one of the eight experimental conditions. All participants first viewed either the trauma or neutral film. They were instructed to immerse themself in the film, and they viewed the film from a projector. They were then exposed to a visual cue reminder before completing either the *Tetris* or the podcast task.

Subsequently, participants completed the vigilance retrieval task (VRT). During the VRT, participants perform a "go/no-go" vigilance task that requires pressing a key whenever a single digit is displayed, except when that digit is a '3'. Blurred scenes, either taken from the films seen previously or foils, appear in the background behind some digits. In the involuntary condition, participants are asked to pause the task (by pressing a key) whenever a "visual image related to the film pops into mind". The voluntary condition is matched to the involuntary condition, except participants will be asked to deliberately recall film content and pause the task whenever they succeed (meeting the "retrieval intentionality criterion") (Schacter & Graf, 1989). Finally, participants completed the post-task valence and arousal ratings for each clip. The two dependent measures that were recorded from the VRT were the total number of spacebar presses, indicative of valid memory reports. (DV1) and the total number of unique details recalled (DV2).

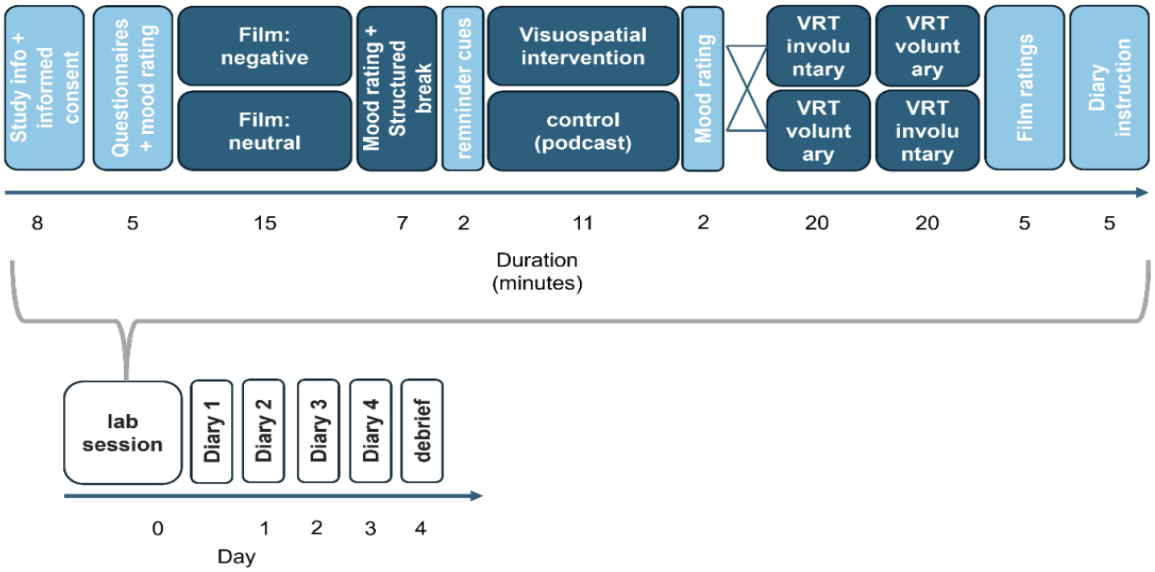


Figure 3: Experimental Timeline (Bernhard et al., 2025)

2.4. Data Processing & NLP Pipeline

Each film clip was further segmented into singular, meaningful chunks, and a description for each clip segment

was written. All memory descriptions (DV1) and film segment descriptions were analysed computationally. First, semantic similarity was assessed using the *all-MiniLM-L6-v2* Sentence-BERT model (Reimers & Gurevych, 2019), producing 384-dimensional embeddings. Memories were matched to the most semantically similar clip, provided cosine similarity exceeded 0.3; otherwise, they were classified as "external details." To examine the overlap between recalled memories and the original film segments, both memory descriptions and film clip descriptions were embedded using the *all-MiniLM-L6-v2* Sentence-BERT model (384 dimensions; Reimers & Gurevych, 2019). These embeddings capture semantic content in a high-dimensional vector space. Cosine similarity was computed between each memory and all film clip embeddings, defined as:

$$Similarity_{i,j} = \frac{\vec{v}_{mem_i} \cdot \vec{v}_{clip_j}}{|\vec{v}_{mem_i}| \, |\vec{v}_{clip_j}|}$$

where i and j denote the embedding vectors of a memory and a clip, respectively. Each memory was assigned to the clip with the highest similarity score, provided it exceeded a similarity threshold of 0.3. Otherwise, it was classified as an *external detail*. To evaluate the accuracy of this semantic matching approach, 15% of the dataset was manually coded.

To quantify emotional features, large language models were applied. Specifically, arousal was estimated by applying the Emotion English DistilRoBERTa-base classifier (Hartmann, 2022) to assign probabilities of discrete emotions, which were weighted by normative arousal values derived from the *Affective Norms for English Words* (ANEW; Bradley & Lang, 1999). Agency was classified using the Hugging Face *language-agency-classifier* (Wan et al., 2023), which probabilistically distinguishes between high-agentic and low-agentic statements. Valence was derived from the Twitter-roBERTa-base sentiment analysis model (Barbieri et al., 2020), scaled on a 1–10 dimension (1 = happiest, 5 = neutral, 10 = saddest). Arousal scores ranged from –1 to 1, and agency scores from 0 to 1. While segmentation was completed by one experimenter, it is important to note that event segmentation itself is a complex process with established methods for experimental validation (Zacks, 2004). Further explorations may involve multiple people to segment clips and then settle on commonly decided segments.

The clip segments were then further classified as either semantic "hotspots", "pre-hotspots", or "others". An arousal hotspot would be the clip segment which had the highest arousal for a specific clip, and a prehotspot would be the segment just before the hotspot. All other segments of the same clip would be categorised as

“others”. A similar scoring system is in place for agency and valence.

For statistical analyses, both frequentist and non-parametric approaches were employed. Firstly, to verify that memories were not always prompted by preceding cues (blurred images in the VRT), cue-memory match percentages were calculated by computing the number of times a memory belonged to the same clip as the preceding cue using a semantic analysis algorithm that the LLM model used to allocate memories to their respective clips. Cue-memory match percentages were analysed using mixed-design ANOVAs. Reaction time distributions to remember a memory after a cue were compared using Kruskal–Wallis tests and Wilcoxon rank-sum tests with multiple-comparisons correction. Finally, logistic regression models tested the predictive value of arousal, valence, agency, and intervention condition on recall intentionality (voluntary vs. involuntary).

3. Results

3.1. Cue-Memory Similarity

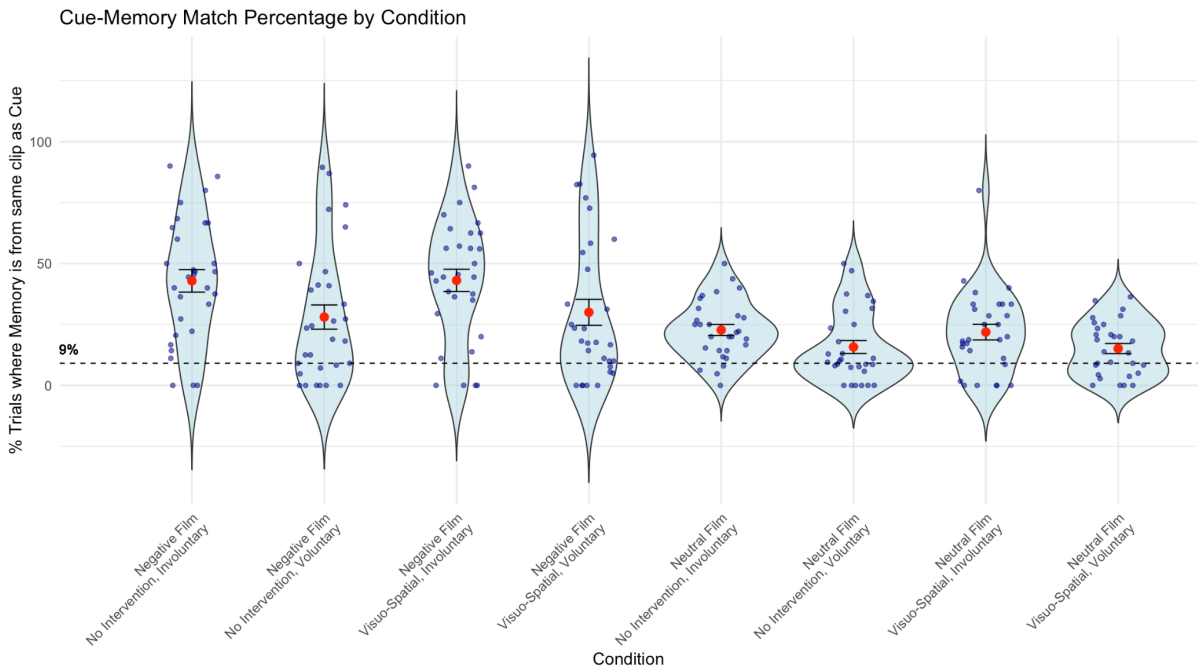
The agreement of the allocation of memories to their clips between the semantic analysis model and human rating is summarised in Table 1. The model achieved high accuracy in both negative and neutral film conditions, with error rates largely attributable to cases in which memory descriptions contained peripheral or ambiguous details. False Negatives were cases in which the memory belonged to a clip, but the model allocated it as an external detail.

Condition	Accuracy	False Negatives*	Wrong Allocation
Negative	87%	3%	10%
Neutral	87%	7%	6%

Table 1: Model accuracy compared to manual coding (15% subset)

Descriptive analysis of the cue–memory similarity was consistently above chance across all conditions (Figure 3). Notably, mean similarity was higher in the negative film conditions compared to the neutral film. In addition, variance was larger for involuntary recall than for voluntary recall, particularly in the negative conditions.

Importantly, a substantial proportion of memories were not directly prompted by the blurred background cues, supporting the feasibility of subsequent hotspot analyses.



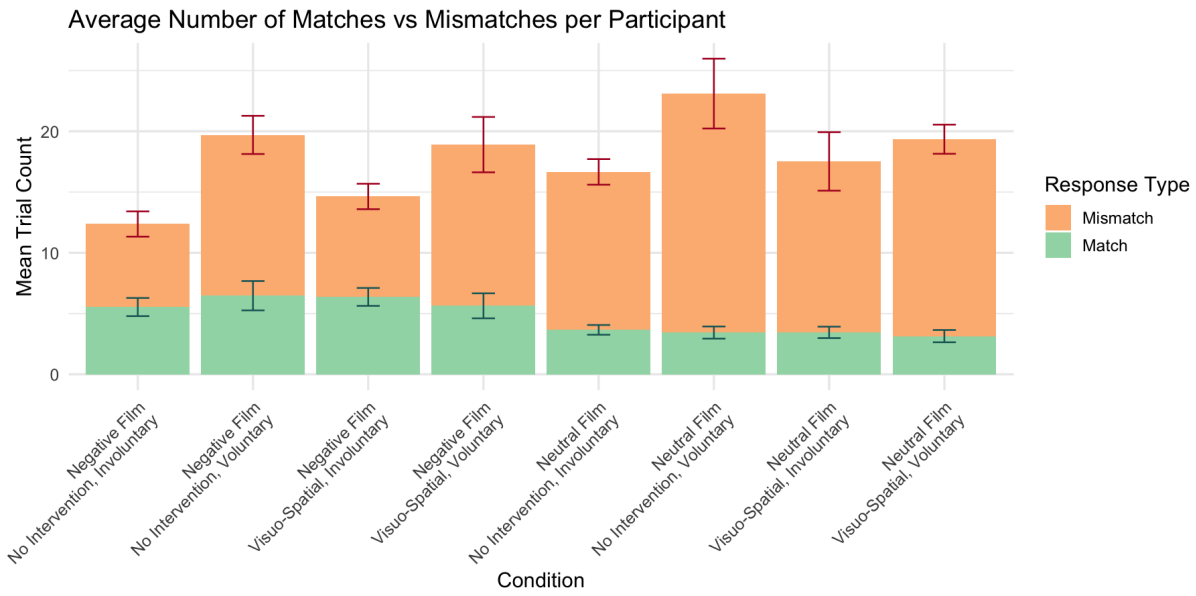


Figure 4.a. (Top) and Figure 4.b. (Bottom):

Cue-Memory Match Percentage by Condition and Average Number of Matches and Mismatches per participant per condition

To be able to conduct a parametric test on this data, the data had to be normally distributed; however, there is an overrepresentation of 0% matches, leading to the data being skewed. 12.66% of participants had 0 matches. To account for skew in the distribution of similarity scores, including overrepresentation of 0% matches, only participants who gave more than 4 memories were considered.

A 2 (Emotion: Negative, Neutral) × 2 (Intervention: Podcast, Visuo-Spatial) × 2 (Intentionality: Involuntary,

Voluntary) ANOVA was conducted on the percentage of matched memories per participant. There was a significant main effect of Emotion ($F_{(1, 229)} = 38.27$, $p < .001$, $\eta^2 = .13$), indicating that participants in the Negative film conditions showed higher matched memory percentages than those in the Neutral film conditions. There was also a significant main effect of Intentionality ($F_{(1, 229)} = 13.80$, $p < .001$, $\eta^2 = .05$) with involuntary retrieval yielding more matched memories than voluntary. The main effect of Intervention ($F_{(1, 229)} = 0.004$, $p = .95$) and all interaction effects (all ps > .20) were non-significant. Figure 5 illustrates the distribution of mean matched memory percentage across the significant effects. Post hoc Tukey-adjusted comparisons indicated that negative involuntary memories were significantly more frequent than neutral involuntary memories ($\Delta = 20.72\%$, $p < .001$, $d = 0.97$), negative voluntary memories ($\Delta = 13.84\%$, $p = .003$, $d = 0.65$), and neutral voluntary memories ($\Delta = 27.42\%$, $p < .001$, $d = 1.28$). Negative voluntary memories were also significantly more frequent than neutral voluntary memories ($\Delta = 13.58\%$, $p = .003$, $d = 0.64$). No significant differences were observed between neutral involuntary and neutral voluntary conditions ($p > .30$).

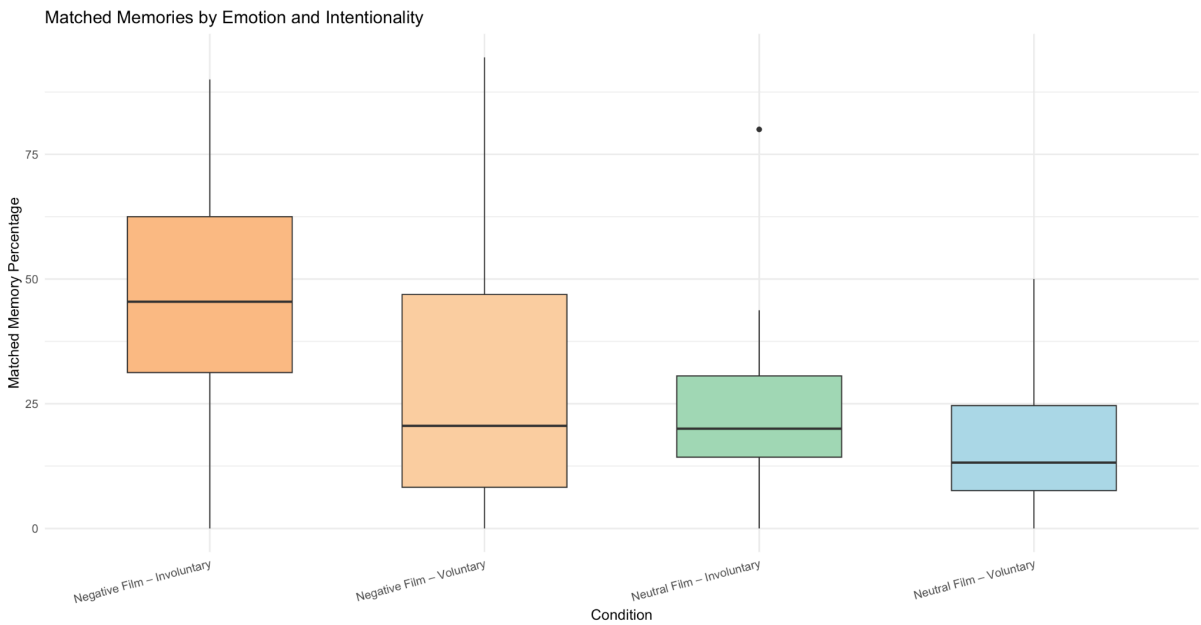


Figure 5: Boxplot comparing the significant predictors for cue-memory matches

The strongest predictor of matched memory was emotional valence, with Negative films consistently producing more matches across conditions. Additionally, involuntary retrieval was associated with more matches than voluntary retrieval, regardless of task or valence. The absence of interaction effects suggests that these influences are additive rather than multiplicative. The time taken to give a matched memory versus an unmatched memory in different conditions was also investigated, and the results are visualised in Figure 6. As

reaction times were not normally distributed, to test whether matched and unmatched memories differed in reaction times within each condition, a Wilcoxon rank-sum test was conducted across the eight experimental conditions. To control for multiple comparisons, p-values were adjusted using the Benjamini–Hochberg method.

A statistically significant difference in summed reaction times between matched and unmatched responses was observed only in the Negative Film, Podcast, Voluntary condition (W = 31,424, p_{adj} = .035). No other conditions showed statistically significant differences after adjustment (all p_{adj} > .10).

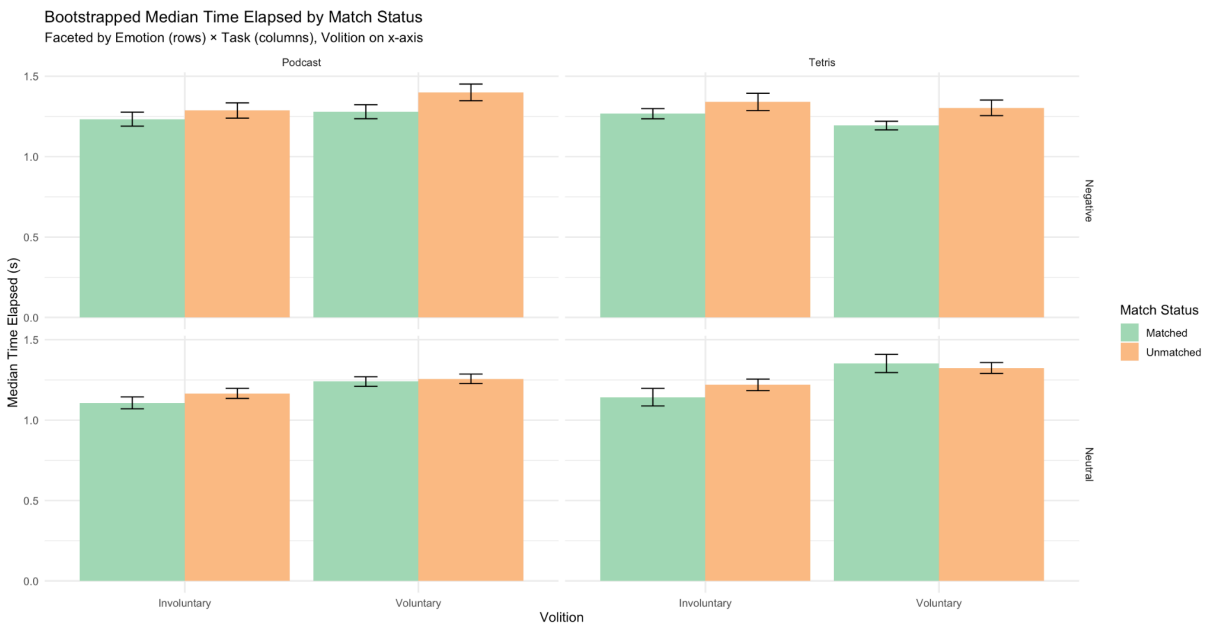


Figure 6: Median Time elapsed between unmatched and matched memories across conditions

3.2. LLM Extraction

Arousal: NLP-derived arousal predictions were positively correlated with participants' post-task arousal ratings, $r(64) = .353$, $p = .003$. Figure 7 displays the correlation scatterplot.

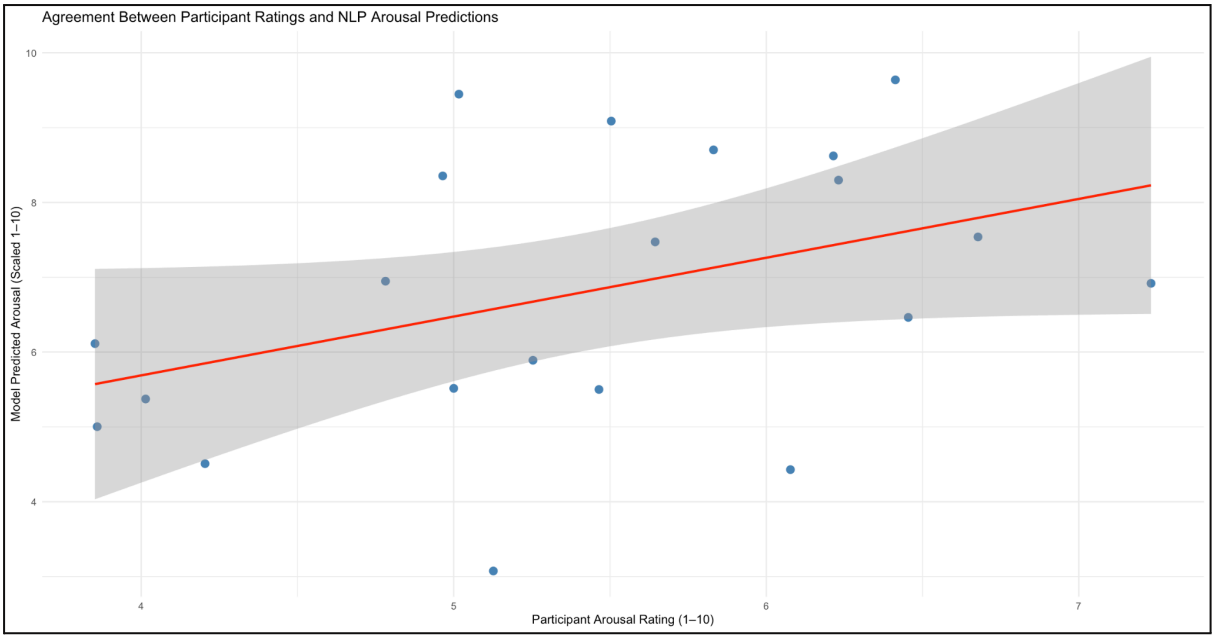


Figure 7: Correlation between participant and NLP model ratings for arousal

Valence: In contrast, NLP valence predictions showed a stronger positive correlation with participant ratings, $r(21) = .687$, $p < .001$, indicating that the model captured subjective valence with higher fidelity (Figure 8). This finding supports the reliability of automated valence extraction from film descriptions and suggests that NLP-based indices can approximate participant emotional experiences.

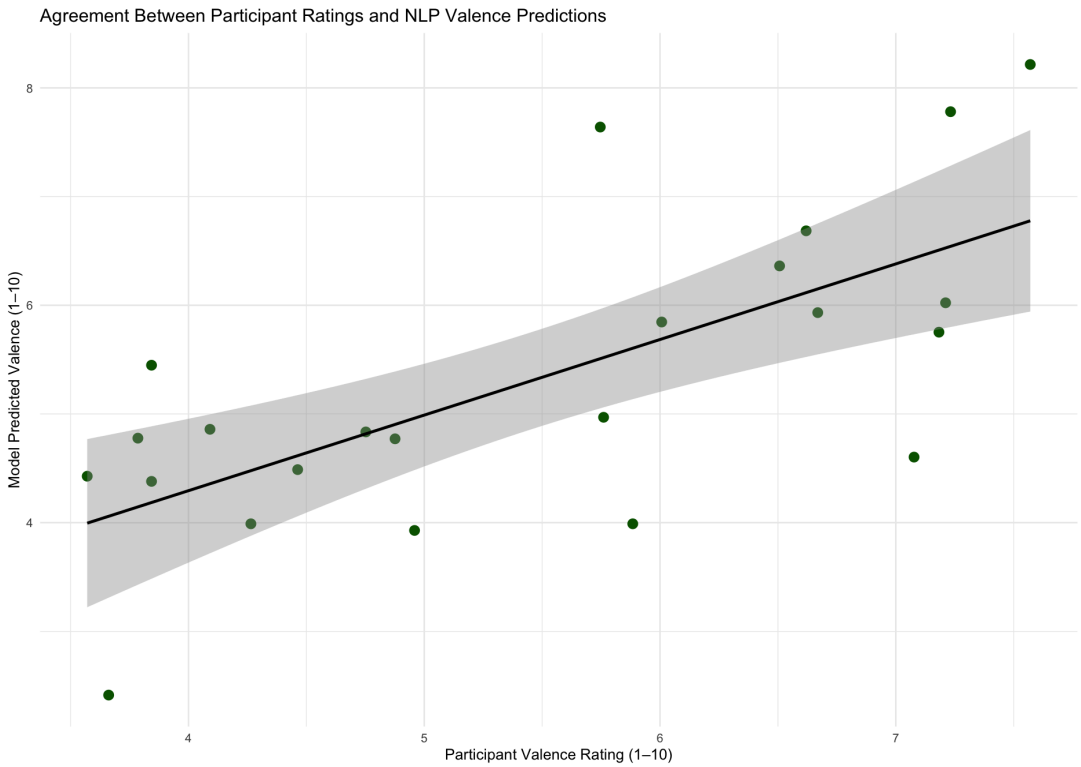


Figure 8: Correlation between participant and NLP model ratings for valence

Distributional analyses: Figure 9 presents violin plots of agency, arousal, and valence distributions across conditions. As expected, arousal scores for neutral clips were centred near 0, while negative clips showed a larger negative magnitude, reflecting stronger emotional arousal. Valence scores clustered around 5 (neutral) in the neutral condition and around 7 in the negative condition, with a left skew in neutral ratings (toward happier values). These distributions align with theoretical predictions and provide convergent validity for the LLM-derived measures. Median agency scores were unexpectedly higher in negative conditions compared to neutral conditions. However, this is likely an artefact of description style (e.g., use of active vs. passive phrasing) rather than a true reflection of stimulus properties. While this raises caution about interpreting agency scores directly, the variability nonetheless highlights potential semantic indicators that may distinguish which memories are more likely to be recalled.

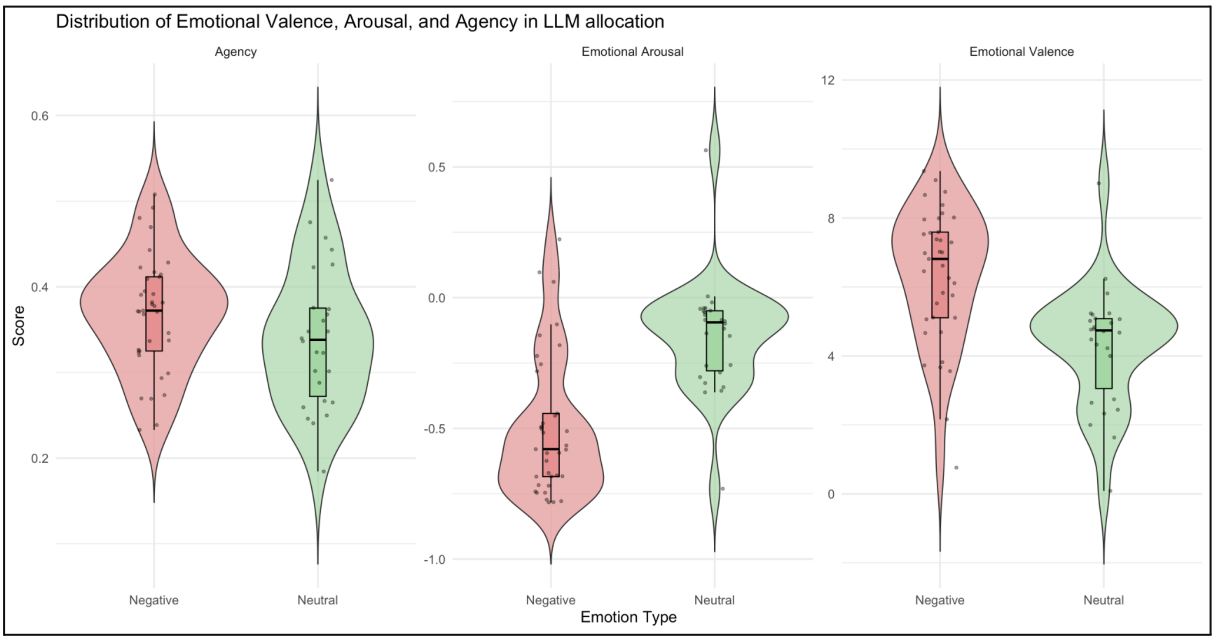


Figure 9: Distribution of NLP extracted semantic ratings of the clip segments

Overall, the LLM models demonstrated promising alignment with human affective ratings, particularly for valence. While the arousal correlation was weaker, the significant relationship suggests the model can still capture broad emotional contrasts. The agency dimension requires more careful validation, as it may be influenced by linguistic artefacts in the text descriptions. Nonetheless, these findings provide initial support for the use of NLP-derived semantic and affective features in modelling trauma film recall, offering a scalable complement to traditional human ratings.

3.3. Descriptive Hotspot Analysis and Logistic Regression

Figures 10, 11, and 12 display the distribution of recalled memories by hotspot category (hotspot, pre-hotspot, other) across agency, arousal, and valence dimensions in the negative condition, separated by recall intentionality (voluntary vs. involuntary) and intervention (Tetris vs. Podcast). The following are descriptive analyses of the hotspots.

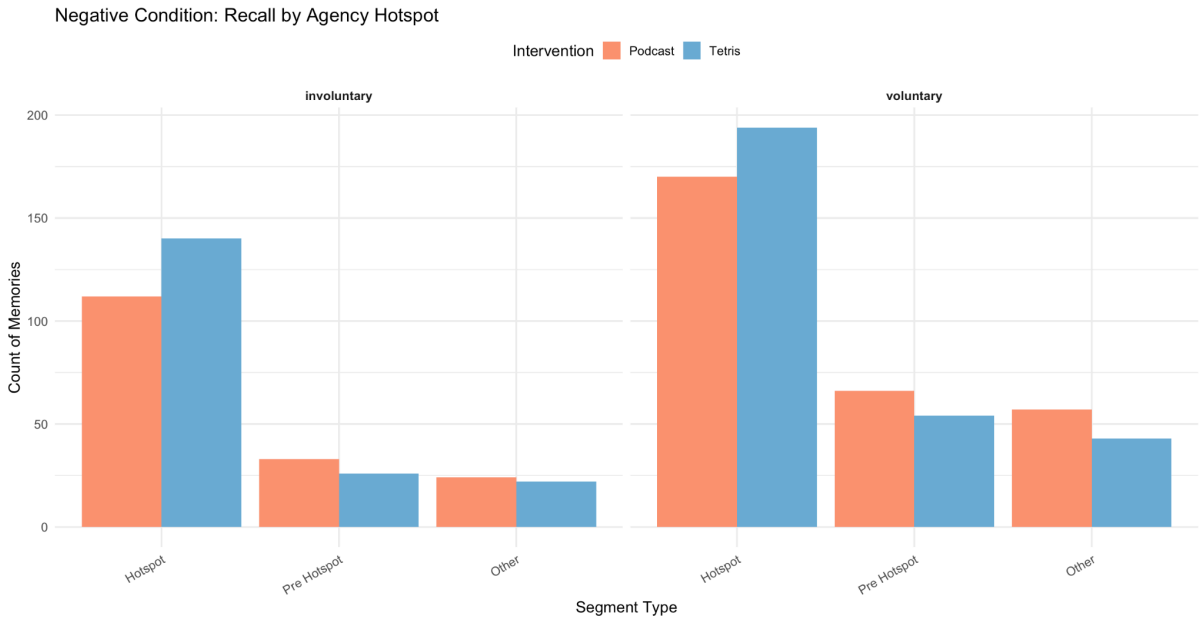


Figure 10: Counts of recalled memories by hotspot type (hotspot, pre-hotspot, other) for Agency in the negative condition. Bars are grouped by intervention (Podcast vs. Tetris) and recall type (voluntary vs. involuntary).

Agency: Hotspot segments were recalled most often, followed by pre-hotspots and others. This pattern held across both voluntary and involuntary recall. Interestingly, recall of agency hotspots was reduced under the Podcast condition compared to Tetris, whereas pre-hotspot recalls increased.

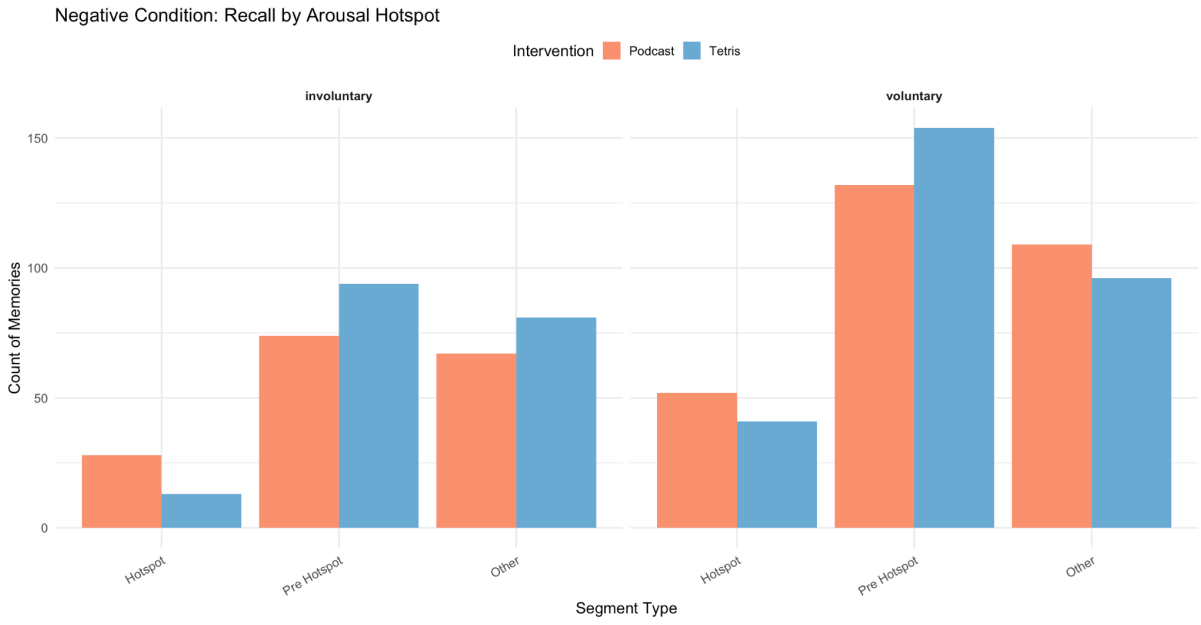


Figure 11: Counts of recalled memories by hotspot type (hotspot, pre-hotspot, other) for arousal in the negative condition. Bars are grouped by intervention (Podcast vs. Tetris) and recall type (voluntary vs. involuntary).

Arousal: Unlike agency, arousal effects showed a reversed pattern: pre-hotspots were recalled more frequently than hotspots, particularly in voluntary recall. Across both recall types, hotspot memories decreased when the intervention was a Podcast rather than Tetris. This suggests that arousal-linked transitions (pre-hotspots) may have heightened salience for memory reactivation. Interestingly, as well, the hotspots were recalled the least often, perhaps evidencing some form of trauma-induced masking.

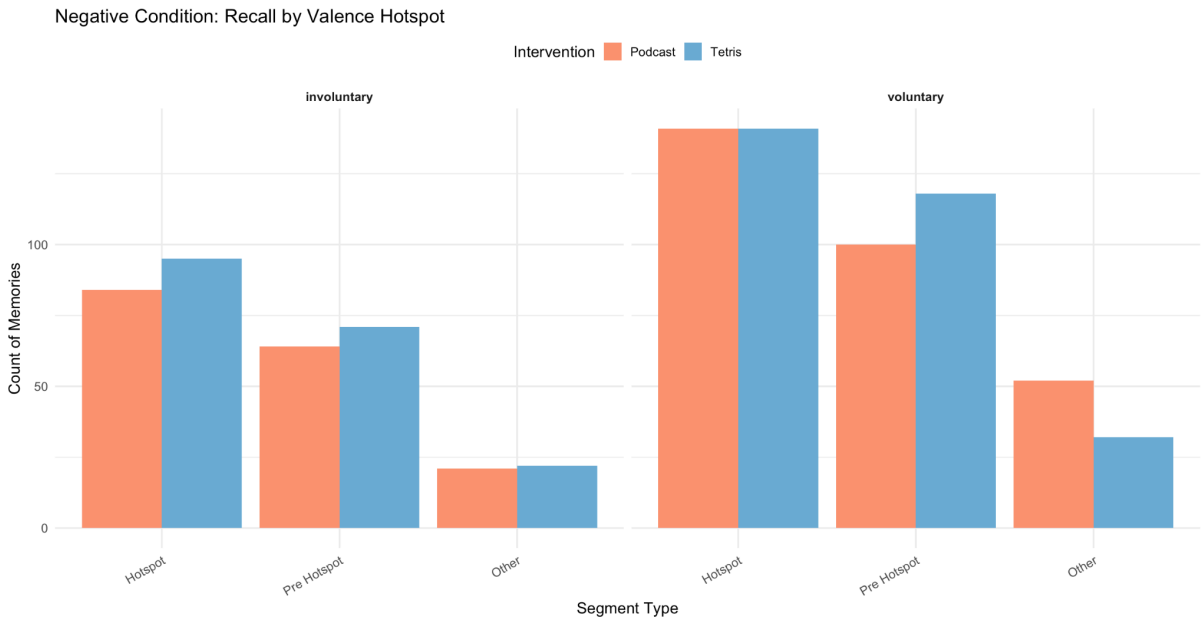


Figure 12: Counts of recalled memories by hotspot type (hotspot, pre-hotspot, other) for valence in the negative condition. Bars are

grouped by intervention (Podcast vs. Tetris) and recall type (voluntary vs. involuntary).

Valence: Valence hotspots were consistently the most frequently recalled segments. However, pre-hotspots were recalled at a substantially higher rate than "other" segments, indicating that emotional build-up preceding the most negative moment of a clip played a notable role in recall.

Across measures, the negative condition showed substantial variability in recall distribution, with pre-hotspots in arousal and valence dimensions disproportionately recalled compared to neutral conditions (see Section 3.2). These descriptive patterns suggest that while hotspots remain important, pre-hotspot segments may play a unique role in shaping involuntary memory intrusions.

Logistic Regression Analyses

To examine the predictors of memory retrieval across conditions, we conducted logistic regression analyses separately for the negative and neutral conditions. This was aimed at noting what the significant semantic qualities of clip segments are that result in their being recalled involuntarily vs voluntarily. The predictors that were tested for the model were visual-spatial interference, arousal, valence, agency, the interaction fo arousal and valence, and arousal, valence, and agency hotspot types (Hotspot, Pre-Hotspot, Other). Figure 13 presents the significant predictors with corresponding odds ratios and 95% confidence intervals as determined by the logistic regression model.

For the neutral model, no predictors reached statistical significance. Although predictors such as arousal showed elevated odds ratios, their wide confidence intervals (all including 1) indicated considerable uncertainty and no reliable effects.

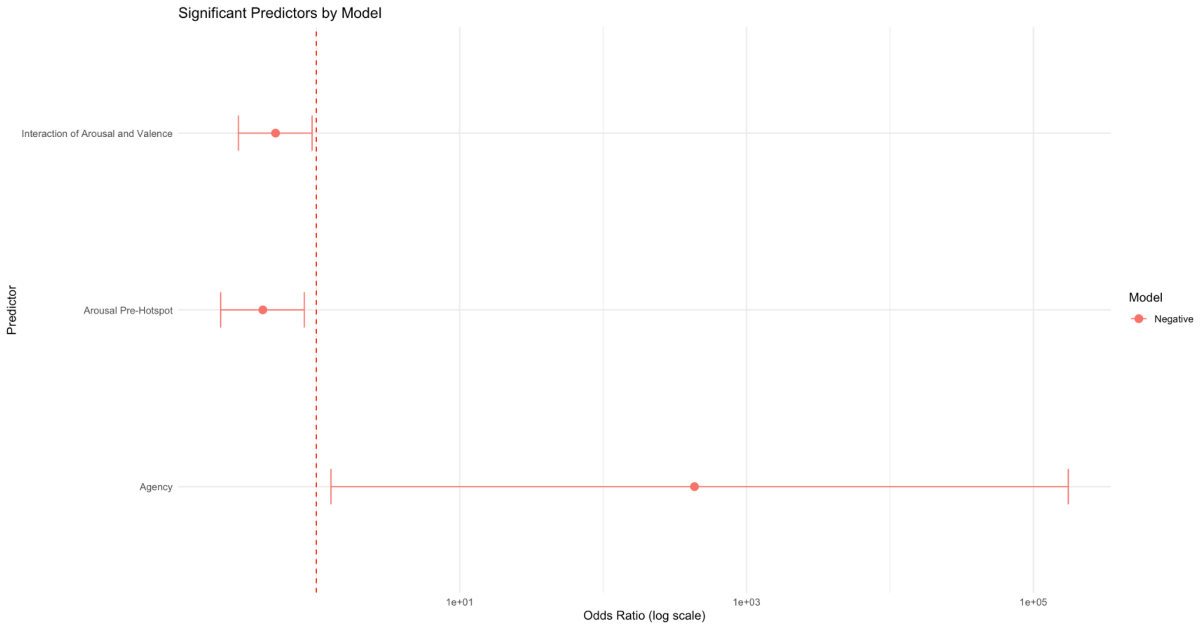


Figure 13: Dot and Whisker Plot of Significant Predictors for the Negative and Neutral Models

For the negative model, three predictors emerged as significant. First, agency was a strong positive predictor of voluntary memory retrieval: higher agency scores were associated with a markedly greater likelihood of voluntary retrieval (*OR* = 433.48, 95% CI [1.27, 174,991.77], *p* = .044). This odds ratio indicates that for each unit increase in agency score, the odds of retrieval increased more than 400-fold (Agency was calculated on a scale from 0 to 1), though the very wide confidence interval suggests caution in interpreting the precise effect size.

Second, arousal immediately preceding the hotspot was negatively associated with retrieval (*OR* = 0.42, 95% CI [0.22, 0.82], *p* = .012). An odds ratio below 1 indicates a decrease in the likelihood of voluntary retrieval: in this case, higher pre-hotspot arousal reduced the odds of voluntary retrieval by about 58% making a segment more likely to be recalled involuntarily. This suggests that memories were more likely to be involuntarily retrieved when arousal was relatively lower just before the most emotionally intense moment.

Finally, the interaction between arousal and valence was significant (*OR* = 0.52, 95% CI [0.29, 0.93], *p* = .030). This indicates that the influence of arousal on retrieval depended on the emotional valence of the material. Specifically, when negative valence was high, increasing arousal further reduced the likelihood of voluntary retrieval. In other words, the combination of high arousal and negative valence made voluntary retrieval less likely, whereas lower arousal in negative contexts was more conducive to voluntary memory retrieval.

Figure 12: Interaction between Arousal and Valence

Figure 14 displays the interaction between arousal and valence for negative memories. Across all levels of valence, higher arousal increased the likelihood of involuntary retrieval. However, this effect was strongest when valence was highly negative: under these conditions, arousal sharply increased the probability of involuntary recall. By contrast, when valence was less negative, arousal had a weaker influence on retrieval type. This pattern illustrates that the combination of strong negative valence and high arousal disproportionately drives involuntary memory retrieval.

Taken together, these findings suggest that for negative material, voluntary memory retrieval was not simply driven by arousal or valence alone. Rather, both the timing of arousal (lower arousal immediately before the most intense moment) and the interaction of arousal and valence played important roles, alongside the perceived sense of agency. In contrast, retrieval of neutral material was not significantly shaped by these factors.

4. Discussion

4.1. Summary of Findings

The present study combined the trauma film paradigm with natural language processing (NLP) methods to investigate what makes certain film segments more likely to be recalled voluntarily or involuntarily. Four central findings emerged.

First, cue–memory similarity was higher for negative films than neutral films, and greater for involuntary than

voluntary recall. Notably, mean similarity was higher in the negative film conditions compared to neutral films, consistent with DRA predictions, and variance was larger for involuntary recall than for voluntary recall, particularly in the negative conditions. This suggests that S-Reps in high-arousal contexts may be more variably reactivated by visual cues than contextualised voluntary memories (Brewin et al., 2014).

Second, arousal strongly predicted involuntary recall. In the negative model, pre-hotspot arousal and the interaction of arousal and valence categories were associated with decreased odds of voluntary recall, implying that these moments were disproportionately recalled involuntarily. These findings highlight arousal as a key dimension of recall intentionality, though its impact is modulated by both timing and valence. Although the correlation between NLP-derived arousal and memory outcomes was statistically significant, the effect size was modest, suggesting that the model was sensitive to broad emotional differences (neutral vs. negative clips) but may have missed perceptual features such as luminance or other visual salience. Furthermore, comparisons with subjective self-reports remain complicated by inter-individual variability in emotional responsiveness (Clark & Ehlers, 2004).

Third, agency-related effects were observed. In the negative model, hotspot agency increased the odds of voluntary recall, suggesting that segments involving action or control during key events were more likely to resurface as voluntary memories. This effect was absent in the neutral model, underlining the importance of emotional context in shaping how agency influences recall.

Finally, the semantic embedding pipeline provided a reliable and theoretically consistent method of mapping memories to film segments. This highlights the methodological contribution of using LLMs to approximate affective features, offering a scalable alternative to manual coding and enabling fine-grained, replicable analyses across much larger datasets than traditional approaches.

4.2. Theoretical Implications

The findings offer several contributions to memory theory, particularly with respect to the Dual Representation Account (DRA). The DRA proposes two parallel traces: sensory-based S-Reps, which are vivid and prone to involuntary triggering, and C-Reps, which rely on contextual binding and are recalled voluntarily (Brewin et al., 2010). Since cue–memory similarity was higher for negative films than neutral films, and greater for involuntary than voluntary recall, this indicates that intrusive memories are more tightly bound to sensory–perceptual cues in negative contexts, consistent with predictions from the Dual Representation Account (Brewin et al., 2010).

Arousal as a driver of involuntary recall. Consistent with the DRA, the results demonstrate that heightened arousal biases memories toward involuntary retrieval. However, the temporal specificity of our findings refines this view. Negative arousal pre-hotspot segments were less likely to be recalled voluntarily, suggesting that anticipatory build-up and peak arousal moments preferentially seed S-Reps rather than C-Reps. This supports clinical observations that individuals often re-experience moments leading up to trauma rather than the traumatic climax itself (Ehlers & Clark, 2000).

Valence–arousal interactions. The significant arousal × valence interaction in the negative model indicates that these dimensions cannot be considered in isolation. For negative clips, higher combined arousal and valence decreased the odds of voluntary recall, suggesting that when emotional meaning is added to affectively charged material, arousal can alter C-Rep encoding. By contrast, in negative contexts, high arousal undermined voluntary recall. This asymmetry suggests that valence shapes whether arousal strengthens contextual encoding or drives intrusive, cue-bound encoding.

Agency and retrieval intentionality. Agency hotspots increased the likelihood of voluntary recall in the negative model, implying that embodied experiences of action or control may be encoded into C-Rep–like traces under emotional stress. This validates theories suggesting that agency supports autobiographical coherence and controllability (Berntsen & Rubin, 2006; Pillemer, 2009). Our findings suggest that in negative contexts, high agency before trauma-related peaks may reduce intrusiveness, perhaps because action-related details are carried out with contextual features in mind and often involve spatial interaction, strengthening C-Reps.

Taken together, these findings suggest that involuntary recall is not explained by arousal alone but emerges from a constellation of features, including arousal timing, valence context, and agency. This multidimensional view offers a more nuanced account of how trauma-related memories form and resurface.

4.3. Methodological Contributions

A major methodological contribution of this work is the demonstration that large language models can reliably approximate affective features such as valence and, to a lesser extent, arousal in naturalistic memory data. Although the effect sizes were modest, the semantic embedding pipeline nonetheless provided a theoretically consistent method of mapping memory reports to film segments. This highlights the translational value of computational approaches: unlike traditional diary or manual coding methods, LLM-based extraction enables the systematic mapping of memory reports back to their originating stimuli at scale, paving the way for larger,

multi-study datasets and cross-laboratory replication. By quantifying cue–memory similarity in a replicable and scalable manner, this work advances methodological tools for studying the specificity of intrusive memories.

4.4. Clinical Implications

First, the disproportionate recall of pre-hotspot arousal segments highlights anticipatory build-up as a target for intervention. Prior research shows that intrusive memories are often preceded by escalating physiological arousal and anticipatory processing, not just the traumatic climax itself (Ehlers & Clark, 2000; Holmes et al., 2004). Therapeutic strategies such as imagery rescripting have traditionally focused on reworking the peak trauma scene (Arntz, 2012), but extending rescripting to pre-hotspot moments where arousal seeds intrusions may be especially effective. Incorporating agency into rescripting has also been shown to reduce distress and enhance perceived control (Morina et al., 2017), suggesting that targeting the anticipatory window with an agency emphasis could yield positive results. This shifts attention from "hotspots," narrowly defined as peak events (Ehlers et al., 2004), to a broader temporal window of vulnerability.

Second, the dissociation between agency-driven voluntary recall and arousal-driven involuntary recall suggests that interventions could be tailored depending on whether the goal is to enhance deliberate remembering or reduce intrusions. DRA (Brewin et al., 1996) highlights the difference between voluntary and involuntary retrieval routes. Enhancing agency in retelling traumatic experiences may strengthen contextual integration and reduce fragmentation (Ehlers & Clark, 2000; Foa & Rothbaum, 1998), whereas techniques that down-regulate anticipatory arousal, such as paced breathing (Zaccaro et al., 2018), may reduce intrusive encoding.

Third, the use of computational NLP models demonstrates a scalable way to identify intrusion-prone moments in patient narratives or therapy transcripts. Computational text analysis has recently been applied to mental health assessment (Althoff et al., 2016) and trauma narratives (Zhang, 2022), suggesting that automated extraction of semantic features such as arousal, valence, and agency could guide clinicians in detecting vulnerable segments. Such approaches could be used retrospectively in trauma accounts (e.g., identifying hotspots in written narratives; Kleim et al., 2013) or prospectively in ongoing exposure therapy.

Beyond theoretical insight, the NLP pipeline itself has translational potential. Automated extraction of intrusion-prone features from patient narratives could support clinicians in targeting interventions more precisely. This aligns with recent calls for scalable, computationally informed approaches in clinical psychology that complement clinician judgement rather than replace it (Young et al., 2023; Chancellor & De Choudhury,

2020).

4.5. Limitations

Several limitations should temper interpretation. First, while the NLP models aligned moderately with participant ratings for arousal and strongly for valence, agency ratings may have been influenced by linguistic artefacts (e.g., active versus passive sentence constructions) rather than genuine psychological features. Additionally, the participant ratings of arousal and valence are taken retroactively, after the VRT was conducted and hence may suffer from unreliable self-report and comparisons that may not reflect actual arousal and valence felt during the moment. It is also important to note that the NLP models themselves do not rely on participant-derived ratings of arousal and valence, but instead on features extracted from the text; alignment with human ratings is therefore correlational rather than built into the modelling process. Caution is warranted in attributing agency effects entirely to cognitive processes. Although our agency measure may have been influenced by linguistic style, the present work demonstrates the feasibility of operationalising agency computationally. Future iterations could combine automated extraction with human coding to validate agency as a meaningful predictor of recall intentionality.

Second, the **ecological validity** of film stimuli is limited. Although the trauma film paradigm reliably elicits intrusions, watching a film is qualitatively different from living through trauma. Real events involve embodied, multisensory, and self-referential processing that films cannot replicate. Thus, generalisation to clinical trauma should be cautious. Third, **statistical power** was adequate for detecting main effects but less robust for interaction terms. The wide confidence intervals, especially for agency odds ratios, suggest that replication with larger samples is needed.

Fourth, **temporal segmentation of clips** was performed by a single experimenter. Event segmentation is subjective, and different segmentations could yield different distributions of hotspots and pre-hotspots. Future work should employ multiple raters or computational segmentation methods to ensure robustness. Finally, while NLP methods offer scalability, they remain limited in sensitivity to visual and non-verbal features that may be crucial for intrusion formation (e.g., brightness, camera perspective, or physiological reactivity).

4.6. Future Directions

First, a closer investigation of pre-hotspot dynamics is warranted. Do anticipatory segments consistently predict intrusions across different emotional contexts and modalities (visual, auditory, somatic)? Longitudinal designs

could test whether such moments predict intrusive diary entries days later. Second, multimodal modelling should integrate semantic features with visual salience, acoustic properties, and physiological measures (e.g., skin conductance). This would provide a richer account of the multisensory encoding that underlies intrusive memories. Third, clinical translation is essential. NLP models could be applied to trauma therapy transcripts or patient journals, with intrusion frequency as a clinical outcome. This would test whether computationally identified hotspots generalise from films to lived trauma. Fourth, interventions could be refined. Beyond Tetris, other visuospatial or attentional tasks might differentially affect hotspot versus pre-hotspot encoding. Testing such interventions may clarify whether intrusion prevention can be targeted more precisely. Finally, future work could examine individual differences. Traits such as baseline anxiety, prior trauma, or narrative style may moderate the relationship between semantic features and recall type. Including such variables would enhance the ecological and clinical relevance of the model.

5. Conclusion

This study investigated the semantic predictors of intrusive versus voluntary memory within the trauma film paradigm, leveraging natural language processing to quantify cue–memory similarity, arousal, valence, and agency. The findings contribute several key insights. First, intrusive memories were more tightly tied to sensory cues in negative contexts, consistent with the Dual Representation Account. Second, arousal, particularly just before hotspots, emerged as a central predictor of involuntary recall, highlighting the role of anticipatory build-up. Third, agency influenced retrieval intentionality, with high-agency segments recalled more voluntarily, suggesting a dissociation between sensory-driven and contextually enriched memory traces.

Together, these results refine current theoretical accounts by demonstrating that intrusiveness is not determined by arousal in isolation, but by the interplay between arousal, valence, and agency, as well as their timing relative to hotspots. They also underscore the potential of computational NLP methods to identify segments most likely to intrude, bridging cognitive theory with applied clinical practice.

Clinically, the ability to predict intrusive-prone moments could help tailor interventions that focus on anticipatory phases of traumatic memories, guiding strategies for prevention and treatment. More broadly, the integration of machine learning with trauma psychology offers a promising new paradigm for scalable, fine-grained analysis of memory processes.

In conclusion, this study advances understanding of the cognitive and emotional mechanisms underlying

intrusive memories and demonstrates the value of computational approaches in trauma research. By identifying the features that make a memory intrusive, it lays the groundwork for both theoretical refinement and novel clinical interventions. As such, it suggests that computational models can help identify memory "hotspots" most likely to intrude, offering potential routes for more precise and effective treatment of trauma-related disorders.

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